Tutorial Worksheet 1

Question 1: Proving Invertibility of the Gram Matrix

Definition. Let A be a matrix with linearly independent columns. Its Gram matrix is $G = A^T A$.

Goal. Show that G is invertible: i.e., that $Gx = 0$ forces $x = 0$. Assume $Gx = 0$, so $A^T Ax = 0$. Left-multiply both sides by x^T:

$$x^T(A^T Ax) = x^T \cdot 0 = 0$$

(a) Recall the transpose rule $(Ax)^T = x^T A^T$. Use this to rewrite $x^T A^T Ax$ as a product of two copies of the same vector. What is $x^T A^T Ax$ equal to, written as $(\ )^T(\ )$?

(b) For any vector v, recall $v^T v = \|v\|^2$. Apply this with $v = Ax$ to your answer from part (a). What does this equal, in terms of a norm?

(c) A vector's squared length is 0 only when the vector itself is the zero vector. What can you now conclude about Ax?

(d) This is the step where the assumption about A finally gets used, since nothing before this point needed it. By definition, columns $a_1, \ldots, a_n$ are linearly independent exactly when $Ax = 0$ has only the trivial solution. Given that A's columns are linearly independent, and $Ax = 0$, what must x be?

Conclusion. We assumed $Gx = 0$ and were forced, step by step, all the way to $x = 0$, which is exactly the definition of invertibility. So $G = A^T A$ **is invertible whenever A's columns are linearly independent.**

Set 1: $a_1 = (1, 2, 1)$, $a_2 = (2, 4, 2)$ **Set 2:** $a_1 = (1, 2, 1)$, $a_2 = (0, 1, 1)$

(e) For each set, check whether $x_1 a_1 + x_2 a_2 = 0$ has a nonzero solution or only $x = 0$, then use this to determine whether $G = A^T A$ is invertible for that set, and identify which step of the proof above breaks down if the columns turn out to be dependent.

Question 2: Least Squares in 3D

A weather station records temperature z as a function of humidity x and pressure y:

x	y	z
0	0	1
1	0	2
0	1	2
1	1	4

We want to fit the plane $z = c + m_1 x + m_2 y$.

(a) Write down the design matrix A, the unknown vector $x^* = (c^*, m_1^*, m_2^*)$, and the target vector b explicitly.

(b) Compute the pseudoinverse $A^+ = (A^T A)^{-1} A^T$, and use it to solve $x^* = A^+ b$.